

G-CAF

GovDigital Composable Architecture Framework

Volume 20

NS Compliance & Foundational Theories

v2.0

Table of Contents

Chapter 1 NS Document Analysis

Original v1.0 analysis: change drivers, monolith findings, modularization

Chapter 2 NS Compliance Summary

6 NS theorems and their G-CAF implementation status

Chapter 3 NS Compliance v1.12

Skills Matrix Standards: artifact hierarchy, change driver analysis

Chapter 4 Business Driver Theory

What business drivers are and how they justify architecture

Chapter 5 Precedence Theory

How to resolve conflicts between competing principles

Chapter 6 Tension Theory

Why principles create tensions and how to manage trade-offs

Chapter 7 Documentation Definition Framework

Schema-driven document assembly from modular building blocks

Chapter 8 Composability Readiness Assessment

92% readiness score across 6 dimensions

Chapter 9 Composability Improvement Plan

Roadmap from 92% to 99% in 3 phases

Chapter 1

NS Document Analysis

The original analysis of G-CAF v1.0 that led to the modular vault design.

Key Finding: G-CAF v1.0 demonstrated strong NS awareness at the architectural level but the document itself exhibited NS violations — multiple change drivers conflated within single sections, creating combinatorial effects when the framework evolved.

Identified Change Drivers

Change Driver	Trigger	Frequency
Governance Structure	Organizational changes (ARB/DDA composition)	Low (yearly)
Security Standards	Threat landscape, compliance requirements	Medium (6–12 months)
Integration Standards	API/event technology evolution	Medium (6–12 months)
PBC Patterns	Composability best practices evolution	Medium (6–12 months)
Data Standards	Data governance policy updates	Low (yearly)
Operational Models	Entity onboarding, model changes	Low (yearly)
Classification Dimensions	New classification needs	Low (yearly)
Technology Standards	Technology lifecycle changes	High (quarterly)

Resolution: decompose the monolithic document into 589 modular files, each with a single change driver, organized by concern in the Obsidian vault structure used today.

Chapter 2

NS Compliance Summary

Normalized Systems ensures bounded changes produce bounded work.

NS Theorem	G-CAF Implementation	Status
Separation of Concerns	3T+2F architecture; orthogonal classification; PBC Types; Domain Models (CD-DDD) separated from	Strong
Action Version Transparency	Interface Contracts; API versioning; stability tiers; Domain Event schemas	Strong
Data Version Transparency	Schema evolution rules (additive-only); tolerant reader mandate; versioned schemas; Aggregate bo	Strong
Separation of State	PBC lifecycle states; Gateway boundaries; clear ownership; Bounded Context boundaries	Strong
Action Instance Traceability	Audit logging; version tracking in Catalog; Domain Event catalog; gateway decision records	Strong
Data Instance Traceability	Audit logging; data lineage (Data Contract); Aggregate-to-Data ownership; data mapping	Strong

Key NS Mechanisms

Mechanism	NS Theorem	Description
3T+2F Architecture	SoC	Each tier/frame has distinct responsibility; change in one tier doesn't cascade
Orthogonal Classification	SoC	Criticality, Sourcing, Stability, PBC Type vary independently
Interface Contracts	AvT	Stable external APIs separate from internal evolution
Schema Evolution Rules	DvT	Additive-only changes; no removal; no rename
Tolerant Reader Pattern	DvT	Consumers must ignore unknown fields; forward compatible
Change Drivers per Module	SoC	Each of 589 modules has a single reason to change
Version Registry	AvT+DvT	Central tracking of all contract and schema versions

Chapter 3

NS Compliance v1.12 — Skills Matrix

Detailed NS analysis of the Skills Matrix Standards release, verifying compliance for each new artifact.

Skills Matrix Artifact Hierarchy

Artifact	Sub-Artifacts	Change Driver
Skill Taxonomy	Domains (TECH, PLAT, DOM, METH), subdomains, registry	Industry, skill evolution
Proficiency Levels	Level definitions (1–5), criteria, assessment methods	Assessment methodology
Role Definitions	Role categories, registry, role levels (L1–L5)	Organizational structure
Role-Skill Mapping	Required skills per role, proficiency requirements	Role evolution
PBC Skill Requirements	Build/Operate/Evolve phase skills, complexity multipliers	PBC multipliers
Skills Gap Analysis	Assessment process, gap severity, remediation planning	Assessment process

Each artifact has its own change driver, ensuring that changes to (e.g.) proficiency assessment methods don't require updating role definitions. NS-compliant by design.

Chapter 4

Business Driver Theory

A Business Driver is a strategic force that necessitates organizational investment and shapes enterprise architecture decisions. Answers: WHY does this principle exist?

Business Strategy → Business Drivers (WHY) → Architecture Principles (WHAT) → Design Patterns (HOW)

Core Principles

Principle	Description
Business-IT Alignment	Principles must trace to business outcomes; without linkage, principles become arbitrary
Stakeholder Value	Each driver represents value for specific stakeholders (Citizens, Government, Operations, Technology)
Measurability	Drivers must connect to measurable outcomes; without metrics, validation is impossible
Completeness	Driver set must collectively justify all principles; unjustified principles indicate gaps

Chapter 5

Precedence Theory

Precedence defines which principle takes priority when two or more create conflicting guidance.

Conflict Scale

With 17 principles, G-CAF has 136 potential conflict pairs $(n(n-1)/2)$. Precedence rules transform ambiguous situations into deterministic decisions.

Principle	Description
Conflict Inevitability	In any comprehensive principle set, conflicts are mathematically inevitable — not a flaw
Context-Dependent Resolution	Same pair may resolve differently: Tier 1 favors resilience; citizen-facing favors UX; PII favors privacy
Explicit Over Implicit	Documented precedence prevents inconsistent decisions, political resolution, post-hoc rationalization
Governance Integration	Rules must integrate with gateway process — not just advisory

Chapter 6

Tension Theory

A Tension exists when two valid principles create opposing guidance in certain contexts. Tensions are trade-offs, not contradictions.

Aspect	Tension	Contradiction
Nature	Trade-off requiring context-dependent choice	Logical conflict — one must be wrong
Both Valid?	Yes, in different contexts	No, one is invalid
Resolution	Define precedence rules per context	Fix/remove the conflicting principle
Example	"Optimize for speed" vs. "optimize for safety"	"Always encrypt" vs. "never encrypt"

Severity Levels

Severity	Definition	Occurrence	Action
High	Frequent conflict; significant impact	Weekly/Monthly	Explicit precedence rule required
Medium	Occasional conflict; moderate impact	Quarterly	Guidance document recommended
Low	Rare conflict; minor impact	Yearly	Document for awareness

Chapter 7

Documentation Definition Framework

Schema-driven approach to assembling G-CAF modules into targeted documentation for different audiences.

Three Layers

Layer	Description	Example
Modules (Source)	Atomic units of content; single concern; versioned if first made	epifirst.mdx, security-contract.md
Building Blocks	Logical groupings by theme/purpose; standard structure	BLOCK-GOVERNANCE, BLOCK-SECURITY
Document Definitions	Audience-specific assembly recipes; output format specific	Executive Briefing, PBC Developer Guide

Modules → Building Blocks → Document Definitions → Assembled Documentation (PDF, HTML, DOCX, PPTX, API)

Chapter 8

Composability Readiness Assessment

Overall Readiness: 92% (HIGH)

Dimension	Score	Status
Frontmatter Coverage	100%	Excellent — all 464 modules have frontmatter
Module ID Coverage	98%	Excellent — 453/464 with module_id
Section Consistency	85%	Good — standard headings (Purpose, Anti-Patterns, Related)
Graph File Coverage	42%	Partial — dependency tracking incomplete
Cross-Reference Density	80%	Good — rich internal linking
Index File Coverage	95%	Excellent — near-complete folder navigation

Chapter 9

Composability Improvement Plan

Objective: increase composability readiness from 92% to 99% in 3 phases.

Phase	Timeline	Focus	Impact
Phase 1: Quick Wins	Weeks 1–2	Add missing module_id to 11 ADR files; fix non-standard format fields	+3%
Phase 2: Standardization	Weeks 3–4	Add _graph.yaml to 49 directories; standardize related_modules format	+2%
Phase 3: Enhancement	Weeks 5–6	Add audience field to 450 files; resolve hardcoded version references	+2%

Priority gap summary: Missing module_id (11 files, HIGH — blocks ID-based assembly), Missing _graph.yaml (49 directories, MEDIUM — limits dependency analysis), Hardcoded version references (~30 instances, MEDIUM — manual update burden), Missing audience field (450 files, LOW — limits audience filtering).

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